

A. FUNCTIONAL PERFORMANCE REQUIREMENTS – GRID INTEGRATION

A.01. INTRODUCTION

This schedule specifies Technical Limits based on preliminary modelling and assessment of the Facility and Network.

In connection with its award of the project, the project proponent will provide NSO with its proposed Technical Limits together with supporting technical information provided by the Construction Developer and equipment supplier(s) as required by the project proponent in accordance with the Specifications (including power systems model in PSS®E and PSCAD format). Any deviations to the Technical Limits herein shall be identified, and demonstration provided that these do not adversely impact the stability, reliability and power quality of the Network.

NSO may request additional data to determine the appropriateness of the Technical Limits.

NSO's feedback on the acceptability of the Company's proposed Technical Limits shall not be unreasonably withheld or delayed and shall be deemed to be given unless NSO informs the Company in writing of any disagreement (giving reasons) within sixty (60) days of receipt by it of the information provided by the Company pursuant to this Schedule.

During the Supply Period, configurable settings for Technical Limits must be reconfigured at the request of NSO to maintain stability following changes in the network. Any such changes shall be agreed with NSO.

The works shall include but not be limited to the following:

- (a) Providing technical limits of the BESS system
- (b) Grid integration and control performance complied with the specification as per this document
- (c) Providing necessary PSS®E and PSCAD models with site specific settings including Releasable User Guides (RUG)
- (d) Power Plant Controller (PPC) controlling point shall be at 33 kV POC (CT & VT point related to this point will be provided by NSO).
- (e) Performance monitoring point shall be at 33 kV (POC)

POC – Point of Connection -33 kV

A.02. DEFINITIONS, STANDARDS AND OPERATING CONDITIONS

The grid forming BESS and associated equipment shall withstand environmental conditions at the Grid substation locations specified by NSO, including wind-blown

dust, rain, and other contaminants. The buildings provided by the developer for the equipment shall be environment controlled and free of contaminants.

A.02.01. Definitions

Unless explicitly stated in this document, or the Contract, definitions are as per the Standards.

A.02.02. Standards

All equipment shall be installed and designed in accordance with statutory requirements. Where explicit local regulations are not applicable, all equipment and works supplied shall conform to the latest editions of the relevant international standards. The following order of precedence shall apply¹:

- (a) Grid Connection Code approved by Public Utilities Commission in Sri Lanka dated 22 July 2024
- (b) Relevant International Electrotechnical Commission (IEC) Standards
- (c) Key standards applicable to the Grid Integration are specified where relevant in the Bidding Documents.

All referenced standards must be complied with unless exception is granted by EMPLOYER. If a Functional Requirement exceeds a standard listed, then that Functional Requirement shall be adhered to.

Standard(s) referred to shall mean the current Edition / Revision together with Amendments issued.

A.02.03. Site and Environmental Data

Refer Clause 1.1.3 of Volume 4 for Site Environmental Conditions.

A.02.04. Temperature and Humidity

The BESS and associated equipment shall be rated for continuous operation and energisation at temperatures ranging from 10 °C to 40 °C. The SVS shall operate in a curtailed mode if the temperature falls outside of the operating range. The developer shall specify such degraded ratings in the proposal.

The design relative humidity shall be 95%, measured over a period of 24 hours.

¹ If similar standards exist in different suites of standards, the lower priority standard need not be applied. Alternatively, the Developer may seek EMPLOYER's approval of application of the lower priority standard without application of the higher priority standard. Such approval shall not unreasonably be withheld.

A.03. CONDITIONS FOR CONNECTION OF BESS

A.03.01. Outline of Requirements

- (a) This schedule sets out details of additional requirements and conditions that the Developer must satisfy as a condition of connection of a generating system or integrated resource system to the power system as specified in the Grid Connection Code. The Developer shall require adhering updated requirements indicated in this schedule which may be slightly deviated from the Grid Connection Code requirements when it specifically indicated in this schedule.
- (b) This schedule also sets out the requirements and conditions which subject to clause 3 of the Grid Connection Code,
- (c) To co-operate with the NSO on technical matters when making assessment; and to provide information to the NSO.
- (d) The equipment associated with each BESS and associated auxiliary equipment must be designed to withstand without damage the range of operating conditions which may arise consistent with the system standards.
- (e) The Developer must comply with the specification given in this schedule, Grid Connection Code, and any attached terms or conditions of agreement agreed with the NSO in accordance with a relevant provision of Grid Connection Code (in line with 3.1 (a)) before hand over the unit to NSO. Clauses in this specification (Technical Requirement) takes precedence over the relevant Grid Connection Code clauses. If any discrepancy, the Developer must clarify it with NSO.
- (f) The Developer must agree with performance criteria with NSO according to the given specification and Grid Connection Code requirement. Once agreed it must be recorded together as the agreed performance standard for evaluation purpose.

A.03.02. Application of settings

- (a) The Developer must only apply settings to a control system or a protection system that are necessary to comply with performance requirements of this schedule. All settings (except internal proprietary settings) must be approved in writing by the NSO. The Developer or equipment manufacturer must not change its setting without prior approval from NSO even though they may have ability to do so.

- (b) If the Developer seeks approval from the NSO to apply or change a setting, then approval must not be withheld unless the NSO reasonably determines that the changed setting would cause the BESS unit to not comply with the relevant performance standard or cause an inter-regional power transfer capability to be reduced.
- (c) If the NSO reasonably determines that a setting of the BESS unit's control system or protection system needs to change to comply with the relevant performance standard or to maintain or restore an inter-regional or intra-regional power transfer capability, the NSO must consult with the Developer to change settings. NSO shall request in writing that a setting be applied in accordance with the determination and the Developer must be able to change those settings
- (d) The Developer who receives such a request must arrange for the notified setting to be applied as requested and for a test to be conducted as requested. After the test, the Developer must, on request, provide NSO with a report of a requested test, including evidence of its success or failure. Such a report of a test is confidential information.
- (e) As mentioned, in A.03.02. (a)-(d) the Developer must not change a setting requested by the NSO without prior written agreement. If the NSO requires the Developer to change a setting within 2 months of a previous request (after commissioning), the NSO must pay the Developer a reasonable cost of changing the setting and conducting the tests as requested.
- (f) There are at least three levels of user access control for NSO staff shall be provided. Finalizing user access control levels shall discuss during the implementation phase.

A.03.03. Technical matters to be coordinated

- (a) The Developer and the NSO must use all reasonable endeavours to agree upon relevant technical matters in connection of the BESS to the NSO network including:
 - i. design at the connection point
 - ii. physical layout of BESS and auxiliary device locations
 - iii. primary protection and backup protection
 - iv. control characteristics as per the specification
 - v. communications facilities as per the specification
 - vi. insulation co-ordination and lightning protection, if required
 - vii. fault levels and fault clearance
 - viii. switching and isolation facilities
 - ix. interlocking and synchronizing arrangements

- x. metering installations
- (b) The Developer must ensure that in designing a generating system's or integrated resource system's electrical plant, including any substation for the connection of the generating system or integrated resource system to the network, to operate at the same nominal voltage as at the connection point:
 - i. the plant complies with the relevant standards mentioned in this schedule and Grid Connection Code unless a provision of the Grid Connection Code allows or requires otherwise
 - ii. the earthing of the plant complies with the IEEE 80: Guide for Safety in AC Substation Grounding
 - iii. the plant is capable of withstanding, without damage the voltage impulse levels specified in this specification (Grid Connection Code Appendix A)
 - iv. the insulation levels of the plant are coordinated with the insulation levels of the network to which the BESS is connected
 - v. safety provisions in respect of the plant comply with requirements applicable to the participating jurisdiction in which the BESS is located, as notified by the NSO.
- (c) If no relevant Standard exists for the purposes of paragraph (b), the Developer must agree with the NSO to comply with another relevant standard.

A.03.04. Provision of information

- (a) The Developer or person who is negotiating a connection agreement with a NSO must promptly on request by NSO all data in relation to that in the Grid Connection Code or this schedule.
- (b) Three months before commissioning of BESS; or
- (c) 5 business days before commissioning of a BESS alteration that is repairing plant after a plant failure, if plant performance after the alteration will differ from performance prior to the plant failure,
- (d) Must provide:
 - i. to NSO
 - i. information about the protections systems of the BESS
 - ii. information about the control systems of the BESS including:
 - i. a set of functional block diagrams, including all functions between feedback signals and generating system output or integrated resource system output or consumption
 - ii. the parameters of each functional block, including all settings, gains, time constants, delays, deadbands and limits
 - iii. the characteristics of non-linear elements

- iv. encrypted models in a form suitable for the software simulation products nominated by NSO and the Grid Connection Code
- (e) The Developer shall provide the service to convert model to another version of PSS®E and PSCAD™ if the provided models are not compatible with the new version structure until the guaranteed lifetime of the unit. NSO shall provide a reasonable cost to the Developer to get the new updated or complied models with a reasonable timeframe which need to be established during tender negotiations.
- (f) The information provided under paragraph (d) must contain sufficient detail for NSO to perform power system simulation studies in accordance with the requirements and circumstances specified in the Grid Connection Code.
- (g) The information provided under paragraph (d) must:
 - i. encompass all control systems that respond to voltage or frequency disturbances on the power system, and which are either integral to the production units or otherwise part of the BESS, including those applying to reactive power equipment that forms part of the generating system or integrated resource system; and
 - ii. conform with the applicable models developed in accordance with the DATA AND MODEL REQUIREMENTS TO BE COMPLIED BY INVERTER BASED RENEWABLE ENERGY PLANTS (IBRE) PRIOR TO BE INTERCONNECTED TO NSO NETWORK (Refer Annex I) , or an alternative model agreed with NSO to be necessary to adequately represent the relevant plant to carry out load flow and dynamic simulations and (where applicable) specialised power system studies.
- (h) The Developer must provide to NSO information that updates the information provided under paragraph (d) and must provide to the NSO information that updates the information provided under subparagraph (b):
 - i. within 3 months after commissioning tests or other tests undertaken in accordance with clause A.05.24. (c) are completed.
 - ii. when the Generator or Integrated Resource Provider becomes aware that the information is incomplete, inaccurate or out of date; or
- (i) On request by NSO, where NSO considers that the information is incomplete, inaccurate or out of date.
 - i. The Developer is only required to provide new information to the extent that it is different to the information previously provided under clause A.05.23.
- (j) For the purposes of BESS design, the technical information that a NSO must, if requested, provide to the Developer in respect of a proposed connection for a generating system or integrated resource system includes:

- i. the highest expected single phase and three phase fault levels at the connection point with the generating system or integrated resource system not connected
 - i. the clearing times of the existing protection systems that would clear a fault at the location at which the new connection would be connected into the existing transmission system or distribution system
 - ii. the expected limits of voltage fluctuation, harmonic voltage distortion and voltage unbalance at the connection point with the generating system or integrated resource system not connected
 - iii. technical information relevant to the connection point with the BESS not synchronised including equivalent source impedance information, sufficient to estimate fault levels, voltage fluctuations, harmonic voltage distortion (for harmonics relevant to the BESS) and voltage unbalance
 - iv. information relating to the performance of the national grid that is reasonably necessary for the Developer to prepare an application to connect, including:
 - a. a model of the power system, including relevant considered projects and the range of expected operating conditions, sufficient to carry out load flow and dynamic simulations; and
 - b. information on inter-regional and intra-regional power transfer capabilities and relevant plant ratings; and
 - v. the NSO expected three phase fault level at the connection point for the BESS following the connection of the NSO.
- (k) All information provided under this clause A.05.23. and A.05.24. must be treated as confidential information.

A.04. POWER SYSTEM CHARACTERISTICS

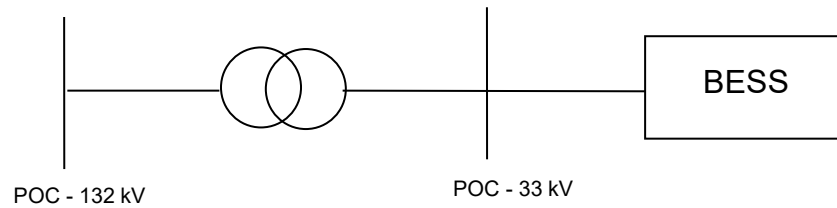


Table A1 – Necessary Power System Characteristics

No.	Type of Information	Value	Units
4	Nominal ac system voltage, line to line at POC	33	kV
5	Maximum continuous ac system voltage, line to line at POC	36	kV
6	Minimum continuous ac system voltage, line to line at POC	30	kV
7	Continuous negative-sequence voltage component (used for performance calculation)	<1 Measured as 10-minute value	%
8	Nominal ac system frequency	50	Hz
9	Maximum continuous ac system frequency (steady state)	50.5	Hz
10	Minimum continuous ac system frequency (steady state)	49.5	Hz
11	Maximum short-term ac system frequency	53	Hz
12	Minimum short-term ac system frequency	47	Hz
13	Maximum rate of change of frequency (df/dt), 500 msec averaging	2.5	Hz/s
17	Basic insulation level at POC	170	kV peak
18	Switching impulse level at POC	-	kV peak
19	Power frequency withstand voltage at POC	70	kV
21	Maximum three-phase fault current at POC (for equipment design)	25 (at 37 kV level)	kA for <u>1</u> s
22	Maximum three-phase fault current at POC	NSO will provide during the tendering process	kA
23	Minimum three-phase fault current POC		kA
26	Maximum single-phase fault current	Not Available	kA
27	Existing single-phase fault current	Not Available	kA
28	Minimum single-phase fault current	Not Available	kA
29	Background harmonic voltage (or current) spectrum (for BESS components rating)	NSO will provide during the tendering process	%
32	Normal fault clearing time at POC	250	ms
33	Breaker failure fault clearing time	550	ms

A.05. TECHNICAL REQUIREMENTS

The offered BESS shall have grid-forming control capabilities, providing short-circuit capacity to the system. During steady-state and transient processes, the grid-forming converter shall instantly share unbalanced grid power, respond to voltage changes, and inject reactive power.

A.05.01. Grid Connection Code

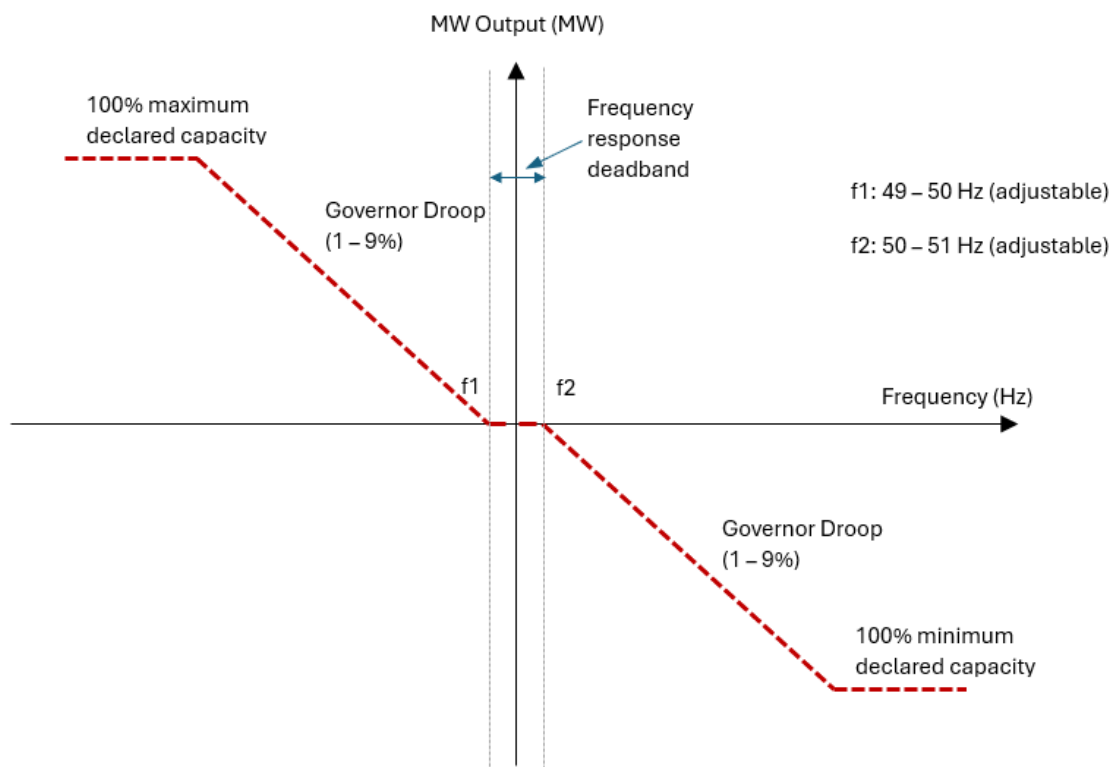
- (a) The connection code applicable for this project is Grid Connection Code approved by Public Utilities Commission in Sri Lanka dated 22 July 2024. However, some clause specifically mentioned in this Technical Requirements may differ from the Grid Connection Code and shall take precedence over the Grid Connection Code requirements. The Developer shall clarify with NSO if there are any concerns on interpretations of the clauses.

A.05.02. Frequency variation capability

- (a) BESS shall be capable of delivering the declared active and reactive power outputs within the system Frequency variations, specified in this specification. Additionally, the PCS AC-side current of the BESS shall be capable of operating continuously at 110% of the rated current, sustain 120% of the rated current for ≥ 2 minutes, and preferably support short-term overloads of 150% rated current for 10 seconds.
- (b) Active and reactive power capability (± 10 MW/ ± 3 Mvar) of the BESS must be available for frequency range between 47 Hz to 52 Hz for at least for time related clause in A.05.04. If wider frequency ranges or longer minimum times for operation are economically and technically feasible, the power park module owner shall not unreasonably withhold consent.
- (c) The Developer shall consider real and reactive power loss transformer when calculating the BESS rating at POC.
- (d) The supplied active power shall be delivered to the network based on the nominated frequency droop settings and deadband settings.
- (e) BESS shall be protected against frequency excursion outside the frequency range specified in clause in A.05.04. and able to disconnect from the system if the extreme frequency could risk to the BESS performance and reliability.

A.05.03. Frequency support capability

- (a) The BESS must be capable of responding independently on a configurable droop curve in response to frequency deviations, without receiving external control signals from NSO.
- (b) Droop set point shall be set based on studies conducted by the NSO as part of the design of the Facility for the purpose of maintaining network stability. Droop setting shall be able to set between 1% to 9% as per the NSO discretion. If NSO did not disclose any Droop value, then it shall be set at 4%.



- (c) During a sudden frequency drop BESS shall mimic an inertia response of a synchronous machine through its grid forming control system. This synthetic inertia response shall be configured by the NSO as part of the design of the Facility for the purpose of maintaining network stability.
- (d) The maximum inertia time constant of the grid-forming electrochemical energy storage system shall be no less than 20 seconds, with an inertia response activation time ≤ 5 ms. The system shall allow flexible adjustment based on grid requirements.
- (e) The response time for primary frequency regulation shall be < 0.2 s, with active power adjustment deviation $\leq 2\%$.

- (f) No frequency regulation limit shall be set in principle, but if necessary, the limit shall be $\geq 20\%$ Pn

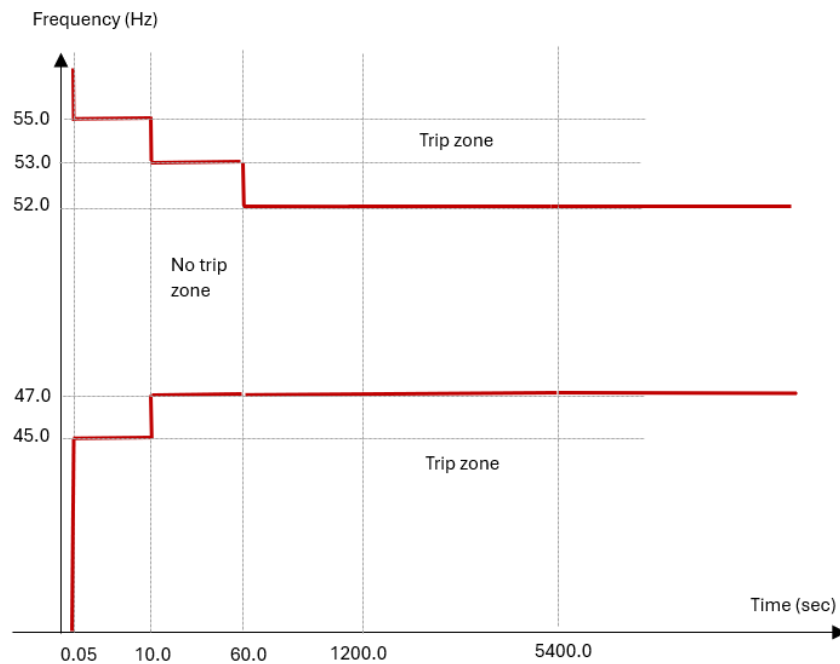
A.05.04. Frequency withstand capability

- (a) BESS shall be capable of staying connected to the Transmission System and remain continuously operable within the System Frequency range 47 Hz to 52 Hz.
- (b) BESS must remain connected and fully operational for under frequency variations between 47 Hz to 50 Hz for duration as shown in the following table and figure. Under frequency variation window shall extend to 45 Hz to 50 Hz if equipment has the capability to do so.
- (c) BESS must remain connected and fully operational for over frequency variations between 50 Hz to 52 Hz for duration as shown in the following table and figure. Over frequency variation window shall extend to 50 Hz to 55 Hz if equipment has the capability to do so.
- (d) Frequency ride through settings shall be configurable in at least four bands between 45 Hz – 55 Hz covering above requirement. If default band range capability of equipment exceeds this requirement, the default capability must be maintained and accessible to NSO without restriction.
- (e) BESS shall change its power level up to its maximum rating according to the droop setting and frequency deviation in the system.

Frequency		Time
Hz	p.u.	
$f > 55.0$	$f > 1.10$	<i>Automatic disconnection allowed if so desired by the BESS Operator after 50 milliseconds</i>
$55.0 \geq f > 53.0$	$1.10 \geq f > 1.06$	10 seconds
$53.0 \geq f > 52.0$	$1.06 \geq f > 1.04$	1 minutes
$52 \geq f \geq 47.0$	$1.04 \geq f \geq 0.94$	Continuous operation
$47.0 > f \geq 45.0$	$0.94 > f \geq 0.9$	10 seconds
$45.0 > f$	$0.90 > f$	<i>Automatic disconnection allowed if so desired by the</i>

Frequency		Time
Hz	p.u.	
		<i>BESS Operator after 50 milliseconds</i>

Above frequency range specified in *Italic letters* shall consider as optional but recommended if the equipment has capability to do so.



A.05.05. Rate of change of frequency

- (a) At a minimum, the BESS shall be able to remain synchronised to the system during rate of change of system frequency of up to and including 4.0 Hz per second measured at a rolling average over 250 milliseconds and 2.5 Hz per second measured at a rolling average over 500 milliseconds.
- (b) Rate of change of frequency (RoCoF) withstand capability mentioned in the A.05.05. (a) does not mean that equipment manufacturer or BESS developer intentionally set the BESS' RoCoF protection to activate when RoCoF limit exceeds above limits in A.05.05. (a) unless violation is a risk to the device functionality or reliability. If any RoCoF protection of the BESS should be disabled unless it is essential to device stability and protection.

A.05.06. Voltage variation capability

- (a) The BESS shall be capable of delivering the declared active and reactive power outputs within the voltage variations of $\pm 10\%$ of the rated voltage at POC
- (b) The BESS shall be capable of delivering the declared rated current during under voltage condition and over voltage conditions as mentioned in section A.05.08.

A.05.07. Voltage support capability

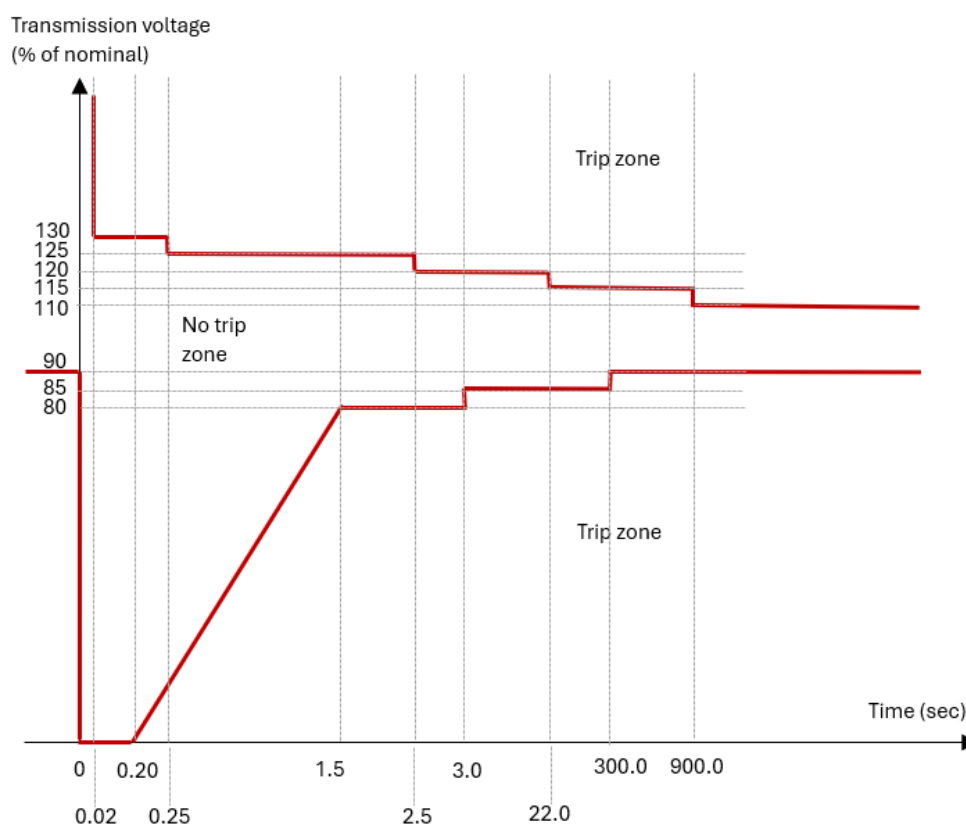
- (a) The BESS must be capable of responding on a configurable droop curve (1 – 6%) to either local voltage deviations or at POC, without receiving external control signals from NSO.
- (b) Droop set point will be set based on studies conducted by the NSO as part of the design of the Facility, for the purpose of maintaining network stability.
- (c) The grid-forming energy storage system shall have automatic voltage regulation (AVR) functionality and the system's 5% voltage step response time shall be < 50 ms, with overshoot $\leq 30\%$ of the voltage step magnitude.

A.05.08. Voltage withstand and fault ride through capability

- (a) Under voltage variations, the BESS must remain connected and fully operational for duration as shown in the following table and corresponding figure.
- (b) Voltage ride through settings must be configurable in at least ten bands between 0 and 1.3p.u. If default capability of equipment exceeds this requirement, the default capability must be maintained and accessible to NSO without restriction.

Voltage at POC	Time
p.u.	
$v > 1.30$	0.02 seconds
$1.30 \geq v > 1.25$	0.25 seconds
$1.25 \geq v > 1.20$	2.5 seconds
$1.20 \geq v > 1.15$	22 seconds
$1.15 \geq v > 1.10$	15 minutes

Voltage at POC	Time
p.u.	
0.90 – 1.10	Continuous operation
$0.90 > v \geq 0.85$	5 minutes
$0.85 > v \geq 0.80$	3.0 seconds
$0.80 > v \geq 0.0$	Linear between 0.2 seconds and 1.5 seconds
$v = 0.00$	0.2 seconds



A.05.09. Response to disturbances following contingency events

(a) BESS must remain in continuous uninterrupted operation for any disturbance caused by:

- a credible contingency event
- a three-phase fault in a transmission system cleared by all relevant primary protection systems (250 millisecond)
- a two phase to ground, phase to phase or phase to ground fault in a transmission system cleared in:

- a. the longest time expected to be taken for a relevant breaker fail protection system to clear the fault; or
 - b. if a protection system referred to in subparagraph (a) is not installed, the greater of the time specified by NSO (or if none is specified, 550 milliseconds) and the longest time expected to be taken for all relevant primary protection systems to clear the fault; or
- iv. a three phase, two phase to ground, phase to phase or phase to ground fault in a distribution network cleared in:
 - a. the longest time expected to be taken for the breaker fail protection system to clear the fault; or
 - b. if a protection system specified by NSO, the greater of 1000 milliseconds and the longest time expected to be taken for all relevant primary protection systems to clear the fault,

provided that the event is not one that would disconnect the production unit from the power system by removing network elements from service.

(b) BESS and each of its production units must remain in continuous uninterrupted operation for a series of up to 6 disturbances within any five-minute period caused by any combination of the events described in paragraph (c) where:

- i. up to four of the disturbances (two LLLG faults, one LL and one LG fault) cause the voltage at the connection point to drop below 50% of normal voltage
- ii. up to one disturbance is cleared by a breaker fail protection system or similar back-up protection system
- iii. up to one disturbance causes the voltage at the connection point to vary within the ranges under clause A.05.08. .
- iv. provided that none of the events would result in:
 - a. the islanding of the generating system or integrated resource system or cause a material reduction in power transfer capability by removing network elements from service
 - b. the cumulative time that voltage at the connection point is lower than 90% of normal voltage exceeding 1,800 milliseconds within any five-minute period
 - c. the time integral, within any five-minute period, of the difference between 90% of normal voltage and the voltage at the connection point when the voltage at the connection point is lower than 90% of normal voltage exceeding 1 pu second.

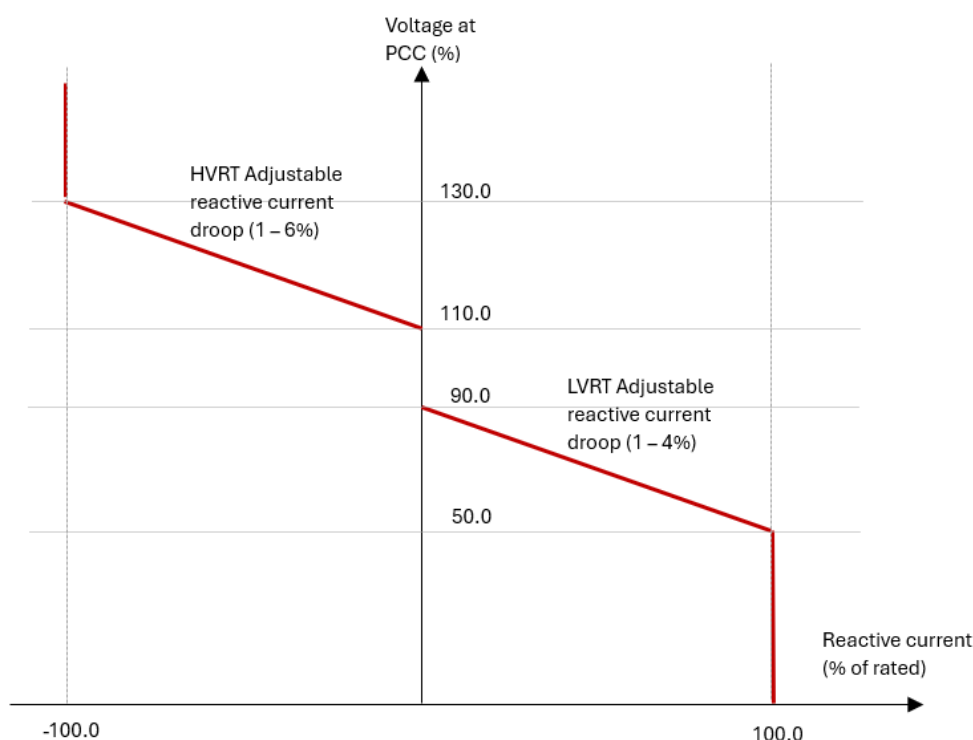
(c) BESS shall provide Inertia support during a fault

- I. The BESS shall provide a synthetic inertia support in response to a fault that mimics an inertia response of a synchronous generator. The maximum inertia time constant of the grid-forming electrochemical energy storage system shall be no less than 20 seconds, with an inertia response activation time ≤ 5 ms. The system shall allow flexible adjustment based on grid requirements.

- II. This synthetic inertia response shall be configured by the NSO as part of the design of the Facility for the purpose of network stability, and subject to approval by NSO.

A.05.10. Reactive current injection and active power recovery during the fault

- (a) Subject to any changed power system conditions or energy source availability beyond the BESS reasonable control, BESS respect of the types of faults described in subparagraphs clause A.05.09. must have BESS capable of supplying to or absorbing reactive current from the network:
- i. to assist the maintenance of power system voltages during the fault:
 - a. BESS inverter shall inject reactive current during the fault (within their capability) based on the amount of remaining system voltage. Amount of reactive current during the contingency shall be equal to 2% for every 1% drop in voltage at the POC as a default value. Further, the BESS unit shall be capable of providing reactive current of x% per 1% drop in voltage at POC where x% shall be set between 1% to 6%.
 - b. Additional reactive current injection during fault shall be in addition to the steady state reactive current given at that time by the BESS.
 - c. during the disturbance and maintained until connection point voltage recovers to between 90% and 110% of normal voltage, or such other range agreed with the NSO; and
 - ii. from 100 milliseconds after clearance of the fault, active power shall recover to at least 95% of the level existing just prior to the fault.
- (b) NSO may request the contribution of positive, negative and zero sequence currents depending on the requirements of fault detection near the POC.
- (c) Allowable activation delays are between 20ms - 40ms (in order for ac protection systems to detect faults without undue delays) and provide or absorb the required reactive current within 70 milliseconds



A.05.11. Quality of electricity generated (harmonic, flicker and voltage unbalance)

- (a) The BESS shall ensure that when connected to NSO network, it must not produce level of distortion or fluctuation of the supply voltage at the connection point:
- i. rapid voltage changes greater than 3% of the nominal voltage during the normal operation
 - ii. harmonic voltage distortion greater than the emission limits allocated by the NSO in accordance to 3.17.2.5 of Grid Connection Code.
 - a. harmonic analysis shall consider harmonic order 2 – 50
 - iii. voltage unbalance greater than the limits allocated by the NSO in accordance with IEC 60034-1.
 - iv. voltage flicker greater than the limits allocated by NSO, i.e., $P_{st} = 0.35$ and $P_{lt} = 0.25$.
- (b) The design of the harmonic mitigation measures (including filters if required) shall be demonstrated through a harmonic analysis study.
- (a) Measurement and evaluation of respective emissions shall be carried out in accordance with IEC 61000-3-6. Measurement and evaluation reports shall be prepared in accordance with IEC 61000-3-6 and IEC 61000-3-7.

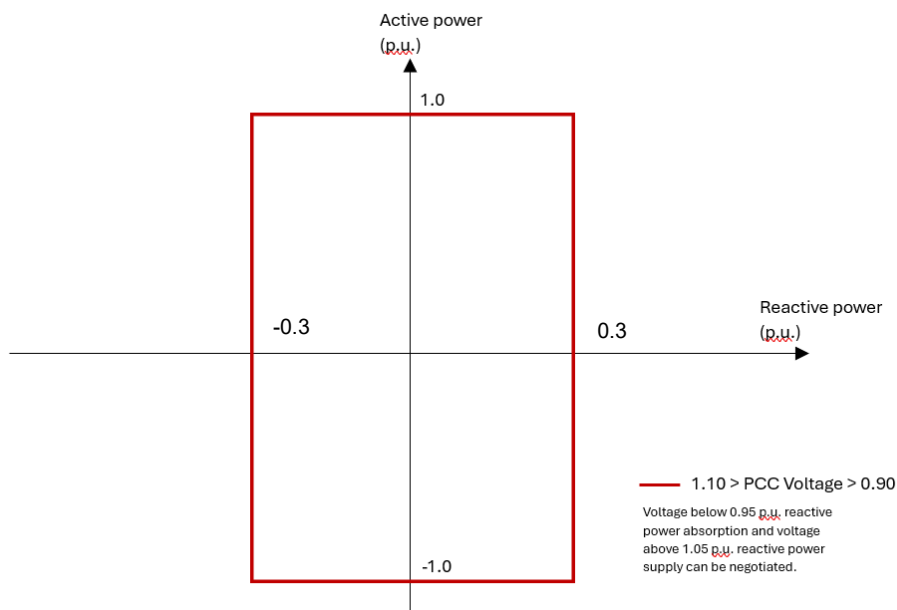
A.05.12. Quality of electricity and continuous uninterrupted operation

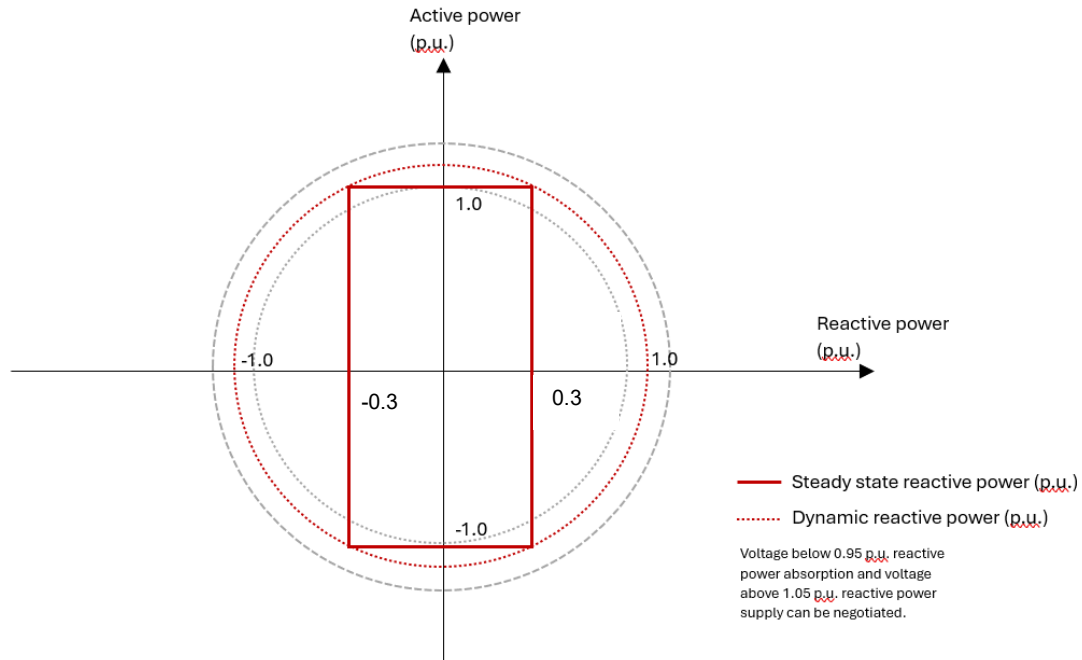
- (b) BESS, harmonic filters and associated reactive plants, if any must not disconnect from the power system as a result of voltage fluctuation, harmonic voltage distortion and voltage unbalance conditions at the connection point

within the levels specified in clauses (as per the planning limits given in 3.17.2.5 of Grid Connection Code.)

A.05.13. Reactive power capability

- (a) The BESS operating at:
 - i. any level of active power; and
 - ii. any voltage at the POC within the limits 0.9 and 1.1 per unit without a contingency event,
- (c) must be capable of supplying and absorbing continuously at its POC (33 kV) an amount of reactive power of at least the amount equal to the product of the rated active power of the BESS and 0.3.
- (d) must be capable to operate in any combination of exporting or importing active and reactive power, otherwise known as four quadrant operation.
- (e) If the BESS cannot meet the reactive power exchange requirements as outlined above, the Developer shall expect to install additional equipment necessary to meet those requirements. The specific selection and design of such equipment shall be approved by the NSO.





A.05.14. Voltage, reactive power and power factor control modes

- (a) The BESS shall be able to provide automatic control at the POC over the entire operating range of the BESS, without causing system instability. Control modes include:
- Voltage control
 - Voltage droop control
 - Reactive power control
 - Power factor control
- (b) The BESS shall be capable of contributing to voltage or voltage droop control at the POC utilizing its capabilities, while respecting reactive power capability and reactive power exchange with the AC network, in accordance with the following control characteristics:
- Voltage set-point at the connection point shall be capable of being specified within the range (0.9 pu-1.1 pu), either continuously or in steps.
 - It shall be possible to operate voltage control with or without a dead band around the setpoint. The dead band shall be adjustable and agreed with NSO.
 - Voltage droop control mode shall include the capability to change reactive power output based on a combination of a modified voltage set-point and an additional instructed reactive power component (i.e. voltage-reactive power droop). The slope (i.e. voltage droop) shall be adjustable in the range (typically 2% -12%).
- (c) The BESS shall be capable of operating at a designated reactive power value specified by the NSO as long as the same is within the reactive power capability of the BESS.

- (d) The BESS shall be capable of controlling the power factor to a target at the POC, while respecting Reactive Power Capability and Reactive Power Exchange with the network.

A.05.15. Set point response

- (a) The BESS inverter must be able to respond independently to separate command set points from National System Control Centre (NSCC) for the following performance characteristics under normal operating conditions:
 - i. MW
 - ii. Mvar
 - iii. power factor
 - iv. voltage
- (b) Set point response shall be by bias of the relevant droop curve and grid support functionality shall be maintained, within the active and reactive power capability limits of the BESS.
- (c) The behavior of the relevant performance characteristic following receipt of a set point command at the Facility SCADA, subject to any pre-set ramp rate limitations, and limitations on the active and reactive power capability limits of the BESS, shall be as follows:
 - i. Set point response must commence within 50 ms.
 - ii. Commanded set point value must be achieved within 300 ms
 - iii. Overshoot of the commanded set point value shall not exceed 10% of the value of the change in set point value.
 - iv. Settling time, to within 1% of full-scale output for the relevant performance characteristic, must be achieved within 400 ms.
 - v. BESS shall automatically notify NSCC when its state of charge moves outside the normal operating state of charge limits as described in Functional Specifications and reduce the active power set point to zero in a manner pre-agreed with NSO.

A.05.16. Priority to Active or Reactive power contribution

- (a) The BESS shall be capable of operating in the active or reactive power priority modes, as requested by NSO, while meeting continuous operating rating and short-term overload ratings. This includes utilizing the full rating (continuous and short-term overload) of the inverter current capability and operating in zero active power transfer. Taking into account the capabilities of the BESS, NSO may inform on case by case basis whether active power contribution or reactive power contribution has the priority during low or high voltage operation.
- (b) If not specified active power control shall have the priority.

A.05.17. Control interaction with other generating devices

- (a) Control Interaction is a specific issue that can impact the coordinated operation of Power Electronic devices in a local area. Control systems of dynamic devices can interact in an undesirable manner resulting in unstable or poorly damped oscillations following system disturbances such as fault recovery.
- (b) The BESS shall design the equipment to avoid such undesirable interactions, and the acceptable operation demonstrated through interconnection studies.
- (c) The BESS shall be capable of contributing to the damping of power oscillations in the connected AC Transmission system. The Developer shall require to provide all required data control system models including parameters for of any small signal modulation controls such as power oscillation damping (POD) controls, or sub-synchronous oscillation damping controls.
- (d) The grid-forming BESS shall have autonomous damping suppression functionality. When low-frequency oscillations (0.2 Hz–2.5 Hz) occur, the system shall suppress oscillations, with active power variation limited to 10%–30% P_n.
- (e) The BESS shall be capable to inject or absorb active phase jump power into the system immediately due to the change of phase angle in the event of disturbance or failure of the system
- (f) Regarding the sub-synchronous torsional interaction (SSTI) damping control, the BESS shall be capable of contributing to electrical damping at torsional oscillation frequencies. The SSTI studies shall be undertaken by the NSO. The studies shall identify the conditions, if any, where SSTI exists and propose any necessary mitigation measures.
- (g) The BESS shall be capable of finding stable operation points with a minimum change in active power flow and voltage level, during and after any planned or unplanned change in the BESS or AC Transmission System to which it is connected.
- (h) The grid-forming BESS shall receive and respond to AGC (Automatic Generation Control) commands from the grid dispatch, with a regulation range of -100% to +100% P_n and steady-state active power deviation $\leq 2\%$ P_n. It shall also receive and respond to AVC (Automatic Voltage Control) commands, with steady-state reactive power deviation $\leq 2\%$ P_n.
- (i) The BESS shall support both grid-following and grid-forming control modes, enabling online switching between the two.

- (j) The grid-following mode shall operate stably under a minimum short-circuit ratio (SCR) of 1.2, while the grid-forming mode shall operate stably under a minimum SCR of 1.0.

A.05.18. Fault current

- (a) the contribution of the BESS to the fault current on the connecting network through its connection point must not exceed the contribution level that will ensure that the total fault current can be safely interrupted by the circuit breakers of the connecting network and safely carried by the connecting network for the duration of the applicable breaker fail protection system fault clearance times, as specified for the relevant connection point by the NSO
- (b) BESS connected plant must be capable of withstanding fault current through the connection point up to the higher of:
 - i. the level specified by NSO as per the clause A.04.
 - ii. the highest level of current at the connection point that can be safely interrupted by the circuit breakers of the connecting network and safely carried by the connecting network for the duration of the applicable breaker fail protection system fault clearance times, as specified by the NSO
 - iii. circuit breaker provided to isolate a generating unit, generating system, bidirectional unit or integrated resource system from the network must be capable of breaking, without damage or restrike, the maximum fault currents that could reasonably be expected to flow through the circuit breaker for any fault in the network or in the generating unit, generating system, bidirectional unit or integrated resource system, as specified in the connection agreement.

A.05.19. Loss of communications

- (a) In the event of loss of communications, the set point response of the BESS shall automatically revert to a state as agreed with NSO during design of the Facility.
- (b) Upon restoration of communications, the BESS plant shall not automatically restore operations but shall assess and issue ready status. Restart of operations shall only be commenced following instruction from NSO.

A.05.20. Safe shutdown

- (a) In the event network fault that leads to power system conditions outside the requirements of this specification, the BESS must automatically execute a safe and controlled shutdown.
- (b) Upon restoration of standard operating conditions, the BESS plant shall not automatically restore operations but shall assess and issue ready status. Restart of operations shall only be commenced following instruction from NSO.

- (c) Subject to the limits of the backup power supply of the Facility, communications and essential functions shall remain operational throughout a safe shutdown event.

A.05.21. Protection from power system disturbances

- (a) BESS to be automatically disconnected from the power system in response to abnormal conditions arising from the power system, the relevant protection system or control system must not disconnect the BESS for:
 - i. conditions for which it must remain in continuous uninterrupted operation or
 - ii. conditions it must withstand under the Grid Connection Code; and
 - iii. a relevant system (as defined in paragraph (a)i) connected to a transmission system must have facilities to automatically and rapidly reduce its generation:
- (b) NSO may require the BESS to be automatically disconnected by a local or remote-control scheme whenever the part of the network to which it is connected has been disconnected from the national grid, forming an island that supplies load.
- (c) Notwithstanding clauses in Grid Connection Code or in this schedule, BESS may be automatically disconnected from the power system under any of the following conditions:
 - a. in accordance with an ancillary services agreement between the BESS and NSO
 - b. where a source of load that is not part of the BESS has the same connection point as the generating system and NSO agree that the disconnection would in effect be under-frequency load shedding
 - c. where the generating system or integrated resource system is automatically disconnected under, Protection systems that impact on power system security or by an emergency frequency control scheme
 - d. where the generating system or integrated resource system is automatically disconnected under Protection to trip plant for unstable operation; or
 - e. in accordance with an agreement between the Developer and NSO (including an agreement in relation to an emergency control scheme) to provide a service that NSO agrees is necessary to maintain or restore power system security in the event of a specified contingency event.

A.05.22. Data and information exchange between BESS and the System Control

- (a) Exchanged Information
 - i. Regarding instrumentation for the operation, the BESS module shall be equipped with an automatic controller capable of receiving instructions from the NSO system control. This automatic controller shall be

- capable of operating the units of the BESS module in a coordinated way.
- ii. The following information shall be provided to NSO by the Developer.
 - a. Typical SCADA signal requirement as per the System Control.
 - b. Detailed high resolution forecast data to be sent to System Control for the PPM having curtailment facility.
 - iii. The BESS controller shall be able to receive the following signals and commands from NSO based on communication with NSO. NSO shall specify the variable type and quality requirements of the supplied signals.
 - a. Start-up command
 - b. Active power order set-point
 - c. Frequency sensitive mode settings
 - d. Reactive power, voltage, or similar set-points
 - e. Reactive power control mode
 - f. Power oscillation damping (POD) control settings, if applicable
 - g. Synthetic inertia, if applicable
 - h. Normal and emergency blocking command
 - i. Ramp blocking command
 - j. Remedial Action Schemes status and settings
 - k. Active power flow direction command, if applicable
 - l. Fast active power reversal command, if applicable
 - m. Redundancy reductions and auxiliary power single source status
 - n. Other project-specific set-points and commands, as specified by NSO.
 - iv. The Developer shall specify any equipment needed to enable the remote selection of control modes and relevant set-points.
 - v. The project proponent shall ensure BESS EMS has the capability to record events where grid support services provided by the BESS and send the same data to NSO periodically.
- (b) The developer of the BESS shall install Grid Code Compliance Monitoring system with following features at the point of Common coupling to ensure that the installed BESS operates in accordance with the Grid Code. The developer should integrate Grid Code Compliance Monitoring system to NSO's central monitoring infrastructure using standardized communication protocols. 3 and should provide comprehensive training for NSO personnel on system operation and data interpretation. Further, the developer shall establish a preventive maintenance program to ensure continuous system reliability.
- i. **Real-time electrical parameter measurement:** The device shall provides continuous monitoring of active power, reactive power, voltage, and frequency with high precision measurement capabilities.
 - ii. **Power quality monitoring:** The device shall be capable of analyzing power quality including harmonic distortion measurement up to the 100th order, voltage unbalance detection, and flicker monitoring.

- iii. **Fault ride-through monitoring:** The device shall detect voltage sag events and validates compliance with low voltage ride-through (LVRT) requirements
- iv. **Communication and control integration:** The device shall support IEC 60870-5-104/101 protocols for seamless integration with NSO's SCADA systems
- v. **Data storage and reporting:** High-resolution data logging and automated compliance reporting capabilities.

A.05.23. Simulation models

- (a) The Developer shall provide all the relevant information details as listed in “Data and Model Requirement to be Complied by Inverter based RE Plants prior to be Interconnected to Transmission Licensees Network”(Annex I).
- (b) The Developer shall provide models that accurately represent the dynamic response of the BESS. The model shall include all site-specific protection, control and other parameter settings as applicable. Both RMS-type and EMT-type models shall be in PSS®E and PSCAD™/EMTDC™ formats, respectively. Further, the Developer shall supply RMS and EMT PowerFactory models to NSONSO. The proprietary information of the BESS that forms the model may be protected. However, the models shall have the flexibility to change parameters and select options that NSO will have access to with the field equipment.
- (c) The PSS®E model shall be capable of running in version 35.x while PSCAD™ model shall be capable of running in Version 5.x with Intel Fortran compiler one API version.
- (d) As a minimum requirement, bidders must fulfill one of the following:
 - I. Submit initial non-site-specific simulation models of the BESS in both PSS®E and PSCAD™/EMTDC™ formats along with the tender for evaluation; or
 - II. provide test results with the tender, demonstrating the BESS response in terms of voltage (V), active power (P), and reactive power (Q) during deep and shallow faults, and for a +50-degree phase angle step change at the Point of Connection (POC), under Short Circuit Ratio (SCR) levels of 1, 3, 5, and 10, with an X/R ratio of 5.

Failure to meet at least one of the above requirements may result in the rejection of the bidder's technical proposal.

- (e) The Developer shall provide the study results in accordance with the requirements of the “Dynamic Plant Model Quality and Dynamic Response Test,” and demonstrate that the plant complies with all functional requirements related to grid forming features specified in this specification. Both the study results and the grid forming demonstrations shall be conducted using PSS®E and PSCAD™/EMTDC™, and submitted for review and

acceptance by NSO. (Refer Annex II). Those simulation studies shall be revised based on the actual system and BESS module tests and adhere to the requirements for the Transmission System and BESS module per the Technical Specifications, as well as the following requirements:

- i. For the purpose of dynamic simulations, the models provided shall contain at least, but not limited to, the following sub-models, depending on the existence of the mentioned components:
 - a. BESS model
 - b. AC component models
 - c. DC system models
 - d. Voltage and power controller (power plant controller)
 - e. Special control features if applicable (e.g. power oscillation damping (POD) function, sub-synchronous torsional interaction (SSTI) control)
 - f. Multi terminal control, if applicable
 - g. BESS protection models as agreed between NSO and the Developer
- (f) The Developer shall verify the models against the results of compliance tests carried out and a report of this verification shall be submitted to NSO within 3 months after the commissioning. The models shall then be used for the purpose of verifying compliance with the requirements of this Regulation including, but not limited to, compliance simulations and used in studies for continuous evaluation in system planning and operation.
- (g) The Developer shall submit BESS recordings to NSO during commissioning, in order to compare the response of the models with these recordings.

A.05.24. Testing, monitoring and compliance review

The additional requirements considered for BESS

- (a) Testing and compliance
 - i. The BESS shall be designed and tested to meet the requirements of the NSO operational requirements, performance studies and any Interconnection Agreements. Sufficient monitoring shall be provided to NSO to determine that the BESS complies with those performance requirements.
- (b) The Developer shall provide commissioning test plan two months before testing dates including number of proposed hold points. Commissioning test plan need to be approved by NSO.
- (c) Grid Connection Testing
 - i. In addition to the site acceptance tests, the Developer shall recommend, provide procedures and carry out compliance tests in line with Best Industry practices. The tests shall include, but not be limited to:
 - a. Harmonic voltage distortion measurements
 - b. Frequency response test

- c. Reactive power compensation test
 - d. Voltage control test
 - e. Tariff metering tests
 - f. Fault ride through capability tests
 - g. Audible Noise measurements
 - h. Performance Guarantee requirements
 - i. Electro-Magnetic Interference measurements
- (d) These tests shall be performed at different active and reactive power levels (worst case also) to demonstrate the features and the stability of the BESS with respect to NSO's AC system.
- (e) The BESS shall include any additional monitoring and recording required for the purpose of completing grid connection testing in the scope of supply.
- (f) The Developer shall coordinate grid connection testing with NSO, including notification to all relevant parties and seeking approval for all test procedures prior the commencement of any system tests.